



Introduction To Programming Using C++ Level 2

Setw Manipulator



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Definition

- Manipulators are tools in C++ used to control the formatting of Input and Output.
- They can be used to control things such as:
 - Field Width
 - Number of Decimal Places
 - Alignment
- ``setw(n)`` means "Set Width" and is used to specify the minimum width of the field in which the next value will be printed.

Explanation

- ◆ `setw(n)` is provided by the `<iomanip>` Header File.
`#include <iomanip>`
- ◆ The basic syntax is:
`cout << setw(n) << value;`
- ◆ `n` specifies the minimum Field Width.
- ◆ `setw(n)` affects only the next value inserted into the Output Stream.
- ◆ If the value is shorter than the specified Width, additional spaces are added to fill the Field.
- ◆ By default, the value is right-aligned, so the spaces are placed before the value.
- ◆ If the value is longer than the specified Width, it is printed completely. `setw()` does not cut or truncate the value.

Procedures

1. Include the `<iomanip>` Header File.
2. Use `setw(n)` before the value you want to format.
3. Specify the desired Field Width inside the parentheses.
4. Use `setw()` again before another value if you want to give that value a different Width.

💡 Example

```
int main()
{
    cout << setw(9) << "C101" << "|"
        << setw(32) << "Introduction to Programming 1" << "|"
        << setw(9) << "95" << "|" << endl;
    return 0;
}
```

➡ Result

C101| Introduction to Programming 1| 95|

Note

- ◆ ``setw()`` affects only the next value inserted into the Stream.
`cout << setw(10) << "Sara" << "Alterkawi";`
- ◆ Only ``"Sara"`` receives the Width of 10.
- ◆ ``"Alterkawi"`` is printed normally.
- ◆ Therefore, if you want to format several values separately, you need to use ``setw()`` before each value.

⊘ Warning

- ◆ `setw()` specifies a minimum Field Width, not a maximum.
- ◆ If the value is longer than the specified Width, the complete value is still printed.
`cout << setw(5) << "Programming";`



Comparison

▨ `setw(n)`

- ✓ Sets the minimum Field Width for the next value.
- ✓ Adds spaces when the value is shorter than the specified Width.
- ✓ Does not truncate values that are longer than the Width.

▤ Without `setw()`

- ✓ Values are printed using their normal Width.
- ✓ No additional spacing is automatically added to create aligned columns.
- ✓ Can make manually printed tables harder to align.

✦ Summary

- ✓ ``setw()`` is a C++ Manipulator used to control Field Width.
- ✓ It requires the ``<iomanip>`` Header File.
- ✓ ``setw(n)`` sets the minimum Width of the next value.
- ✓ By default, values are right-aligned.
- ✓ If the value is shorter than the Width, spaces are added before it.
- ✓ If the value is longer than the Width, it is printed completely.
- ✓ ``setw()`` affects only the next value, so it must be used again for each value that needs a specific Width.
- ✓ ``setw()`` is especially useful for formatting Tables and making Console Output easier to read.

Thank you 😊